Supplemental Information S1: Measurement-Dependent Clustering in the Growing Urban Observatory Sensor Network

Ammar Arbaaeen[2], **Wael Alnahari**[1], **Ammar Alsaig**[2], **Tawfiq Hasanin**[1], **and Wasem Alnahri**[3]

[1]**Department of Information Systems, King Abdulaziz University, Jeddah 21589, Saudi Arabia**
[2]**Department of Computer Science, Umm Al-Qura University, Mecca 24381, Saudi Arabia**
[3]**Independent Researcher, Mecca 24331, Saudi Arabia**

Corresponding author:

Wael Alnahari[1]

Email address: walnahari0004@stu.kau.edu.sa

ABSTRACT

Supplemental Information S1 for the manuscript. This document carries the robustness forks, sensitivity sweeps and estimator controls that the main text summarises but does not tabulate. Sections and tables are numbered with an S prefix so the main-text pointers resolve against this document.

This supplement carries the robustness forks, sensitivity sweeps and estimator controls that the main text summarises but does not tabulate. Sections are numbered with an S prefix, and tables with the same, so a main-text pointer resolves against this document without translation. Section S1 gives the trend tests under every interval procedure; S2 every procedure named in the main text with the value it returns on the headline series; S3 the rarefaction and its anchor sensitivity; S4 the multiscale estimator and its known-answer controls; S5 the reference-frame and grid sweeps; S6 the edge-corrected clustering, the `cdf` correction and the endpoint check; S7 the theme-stratified trend; S8 the gap-filled sensitivity for the multiscale function and the infill test; S9 the infill diagnostics; S10 the snapshot audit in full; S11 the clustering control as a draw and the leave-one-out intensity check; S12 the snapshot's duplicate registrations and two exclusion sweeps; S13 what the snapshot actually contains by ingestion feed; S14 the core-definition sensitivity; S15 the grid-origin sensitivity for the concentration index; S16 the independent cross-check against spatstat; S17 the thinning-invariance test; S18 the near-duplicate test that qualifies the headline decline; and S19 the multiplicity count, the per-year-pair clustering test with its false-discovery correction, the seed check and the code inventory. Every table is generated from the analysis result files by `make_supplement.py`, so the numbers here and in the main text share one source.

S1 TREND TESTS UNDER EACH INTERVAL PROCEDURE

The main text rests the fixed-hull decline on the three tests that are correctly sized at fourteen points. The HAC family is reported here, with its own size. Simulating the OLS-plus-Newey–West statistic at $n = 14$ on pure noise with a $t(12)$ critical value rejects at a nominal 5% about 16.3% of the time, and about 19.3% of the time at the lag-1 autocorrelation the headline series actually shows, so the test is anti-conservative at this sample size. That is why it is not the primary evidence.

Table S1. The clustering trend on the fixed hull under each procedure. The OLS and permutation p-columns are nominal.

Series	Slope/yr	OLS	Perm.	M–K	NW interval	n
Hull R, locations	-0.0107	0.037	0.038	0.010	[-0.019, -0.003]	14
Hull R, identifiers	-0.0126	0.018	0.020	0.010	[-0.021, -0.004]	14
Hull R, locations, no 2026	-0.0096	0.095	0.101	0.030	[-0.020, +0.000]	13
Hull R, locations, no 2017–18, 2026	-0.0096	0.072	0.075	0.017	[-0.019, -0.001]	11

Table S2. Size of the OLS-plus-Newey–West test at $n = 14$ with a $t(12)$ critical value, simulated on series with no trend. A correctly sized test rejects at the nominal rate.

Error process	Newey–West rejects	OLS rejects
AR(1) $\rho = 0$	0.163	0.049
AR(1) $\rho = 0.15$	0.193	0.085
AR(1) $\rho = 0.3$	0.237	0.128
AR(1) $\rho = 0.5$	0.294	0.217
i.i.d. Gaussian	—	0.047

S2 EVERY PROCEDURE NAMED IN THE TEXT, WITH ITS VALUE

The main text names a family of inference procedures and reports the intervals they give. This table puts them side by side on the series that carry the result, so that a reader can see which procedure is responsible for which claim, and can tell a disagreement between procedures from a disagreement between studies. The size simulation at the foot is the check that the HAC interval is not merely narrower than the others: at fourteen points neither the OLS nor the HAC test attains its nominal size.

Table S3. Every inference procedure named in the main text, on the headline series and the rarefied one. "Contains 0" records whether the interval spans zero.

Procedure	Hull R, locs	Hull R, ids	Rarefied R	Contains 0
OLS slope	-0.01071	-0.01257	+0.00031	no / no / yes
OLS interval (t_{12})	[-0.0207, -0.0008]	[-0.0226, -0.0025]	[-0.0047, +0.0053]	no / no / yes
Permutation	0.038	0.020	—	—
Mann–Kendall	0.010	0.010	—	—
Newey–West (HAC, lag 2)	[-0.0188, -0.0027]	[-0.0213, -0.0039]	[-0.0045, +0.0052]	no / no / yes
Theil–Sen	-0.00955	-0.01461	-0.00017	no / no / yes
Theil–Sen interval	[-0.0217, -0.0047]	[-0.0278, -0.0047]	[-0.0045, +0.0062]	no / no / yes
Moving-block bootstrap (box)	[-0.578, -0.305]	—	—	no
Prais–Winsten	-0.0754	—	—	no
Fixed-b (Kiefer–Vogelsang)	[-0.1017, -0.0509]	—	—	no
TOST, full margin	$p = 1.8 \times 10^{-9}$	—	—	equivalent
TOST, half margin	$p = 3.9 \times 10^{-6}$	—	—	equivalent
Power at 80 %	0.99999 for a change of the raw size			—
Size of the 5 % test (simulated)	$\rho = 0$: OLS 0.049, HAC 0.163; $\rho = 0.5$: OLS 0.217, HAC 0.294			—

Every procedure agrees in sign. The OLS test is correctly sized in the same simulation, rejecting 4.7 % under independent errors, but it is mildly optimistic once the errors are autocorrelated: 8.5 % at the lag-1 residual autocorrelation the fixed-hull series has (+0.15) and 21.8 % at $\rho = 0.5$. The HAC test is anti-conservative throughout, 16.3 % under independence and 29.4 % at $\rho = 0.5$, so it is not a safe substitute for the OLS test at this length. Because no single p-value here is calibrated for every error process, the main text carries each claim on more than one procedure and reports the split rather than the strongest interval.

S3 RAREFACTION AND THE ANCHOR SENSITIVITY

Rarefying every year to a common sample size removes the trend in the clustering index and in the concentration index alike. Table S4 is the anchor test: the anchor year's rarefied value equals its raw value by construction and therefore carries unusual leverage, and dropping it shows whether the residual slope belongs to the rarefaction or to the anchor. Table S6 states the scale the rarefied index can resolve, which is the bound on what the main text may conclude from it.

Table S4. Rarefied concentration index and the anchor sensitivity.

Series	Slope/yr	OLS p	Perm. p	M–K p
Rarefied Gini, all fourteen years	-0.0043	0.059	0.059	0.036
Rarefied Gini, 2013 anchor dropped	-0.0026	0.246	0.247	0.129
Rarefied Gini, 2013 and 2026 dropped	-0.0023	0.383	0.377	0.197

Table S5. The rarefied clustering index year by year, with the unrarefied series for comparison. Neither rarefied series has a trend.

Year	Rarefied R	Unrarefied R	Difference
2013	0.357	0.357	0.000
2014	0.319	0.316	0.002
2015	0.388	0.279	0.109
2016	0.356	0.313	0.043
2017	0.369	0.233	0.137
2018	0.440	0.425	0.015
2019	0.414	0.331	0.083
2020	0.356	0.288	0.068
2021	0.461	0.407	0.054
2022	0.369	0.285	0.084
2023	0.364	0.216	0.149
2024	0.373	0.192	0.181
2025	0.344	0.152	0.192
2026	0.352	0.155	0.196

Table S6. The scale the rarefied index can resolve. At the smallest year's sample size the expected nearest-neighbour distance under the reference density is a substantial fraction of the network's own extent.

Quantity	Value
Rarefaction sample size (identifiers)	41
Hull area (km^2)	4416
Expected nearest-neighbour distance (m)	5189
Location sample size, smallest year	39
Anchor sizes tested for the concentration index	39, 150, 200

S4 THE MULTISCALE ESTIMATOR AND ITS CONTROLS

The multiscale function is computed on the convex hull of the full record, and a hull is not a rectangle. A translational edge weight $(1 - d_x/a)(1 - d_y/b)$ is valid only for a rectangular window, and applied to a hull it biases the curve. Table S7 is the known-answer control for the estimator's own convention: on a complete-spatial-randomness pattern, where L is zero by definition, the rectangular weight returns a large spurious value while the border correction returns a value near zero. "Near" is not "zero", and the difference is the subject of Section S11: the control is itself a draw, so it carries sampling scatter, and the residual left after averaging is the bias of the estimator. Over 199 realisations that residual has a

mean absolute value of 262.7 m at $N = 39$, 33.3 m at $N = 200$ and 4.1 m at $N = 561$ (Table S22). The correction is therefore adequate at the sample sizes of the later years and weakly biased in the earliest, sparsest year, which is why the multiscale claims are made for the scale range where the separation is unanimous and not for every scale examined.

Table S7. Known-answer control for the multiscale estimator. A complete-spatial-randomness pattern inside the hull must return $L = 0$ at every scale.

Pattern	100 m	250 m	500 m	750 m	1000 m	1500 m	2000 m
CSR, border correction	+76.7	+104.0	+4.9	+11.3	-5.1	+130.9	+92.0
CSR, rectangular weight	+25.2	+0.4	-146.0	-218.9	-291.9	-345.9	-540.2
Clustered	+1605.7	+3517.3	+4784.3	+4793.0	+4553.5	+4053.5	+3553.5
Regular lattice	-100.0	-250.0	-500.0	-750.0	-1000.0	-1500.0	-2000.0

Table S8. The multiscale function against an inhomogeneous null, on deduplicated locations. Each cell counts years, of fourteen, in each position.

Scale	Above null	Inside	Below
100.0 m	13	1	0
250.0 m	14	0	0
500.0 m	14	0	0
750.0 m	14	0	0
1000.0 m	14	0	0
1500.0 m	14	0	0
2000.0 m	12	2	0

Table S9. Curve shape, each year's multiscale function normalised by its own 2 km value so that the window's effect on the level cancels.

Year	100 m	250 m	500 m	750 m	1000 m	1500 m	2000 m
2013	+0.12	+0.32	+0.57	+0.73	+0.90	+0.96	+1.00
2017	+0.22	+0.35	+0.49	+0.58	+0.67	+0.85	+1.00
2020	+0.19	+0.31	+0.45	+0.57	+0.68	+0.85	+1.00
2023	+0.26	+0.35	+0.47	+0.56	+0.65	+0.83	+1.00
2026	+0.21	+0.26	+0.36	+0.46	+0.56	+0.78	+1.00

S5 REFERENCE FRAMES, UNITS AND THE RAREFACTION GRID

The clustering index is sensitive to the area in its denominator and the concentration index to the grid. The main text reports the headline frame and unit; the sweep is here.

Table S10. Rarefied concentration index at four grid resolutions. Rarefaction reverses the sign of the slope rather than only flattening the series.

Cells per side	Rarefied slope	Theil–Sen interval	Uncorrected slope	Cell (m)
20	-0.0043	[-0.0086, -0.0008]	+0.0128	4897
30	-0.0068	[-0.0133, -0.0034]	+0.0108	3265
40	-0.0106	[-0.0164, -0.0035]	+0.0086	2449
50	-0.0076	[-0.0129, -0.0028]	+0.0108	1959

S6 EDGE-CORRECTED CLUSTERING AND THE 2026 ENDPOINT

The headline Clark–Evans R is uncorrected for edge bias, and in a longitudinal study both N and the spatial extent move, so edge bias could itself trend. Table S11 gives four edge estimators on the fixed

hull and S12 their trends, and the trend table adds a fifth computed with an independent implementation. The first is the headline estimator with no correction; the second keeps only points strictly inside the hull, which drops the handful of sites lying exactly on it; the third is Ripley's border correction (Ripley, 1977), using each point's own nearest-neighbour distance as its guard width; the fourth uses a fixed 500 m guard area. All five decline, with slopes between -0.0107 and -0.0159 per year, so edge bias does not drive the trend. The border estimator carries its own small bias: on complete-spatial-randomness controls it returns 0.90 at $n = 100$ and 0.98 at $n = 700$, so it is reported as a sensitivity and not as the headline.

A fifth estimator is in Table S12 because it is the only one computed with an independent implementation. Donnelly's correction (Donnelly, 1978) is written in terms of the window's area and perimeter, both of which a hull has, but spatstat (Baddeley and Turner, 2005) refuses it on any non-rectangular window, reporting that it is *only available for rectangular windows*, because Donnelly's coefficients were estimated by simulation on rectangles. Its `cdf` correction makes no for random points drawn in the *same* window. On our hull and points that correction gives a 2013–2026 fall of 57.2 % against 45.0 % for the raw index, and a steeper fitted slope, so the general-region correction steepens the uncorrected series. It does not touch the question the tolerance test raises, because co-location is a question about which records are one unit rather than about the window edge. On complete-spatial-randomness controls in the same hull it returns 0.992 at $n = 500$ where the raw index returns 1.015, so it is the better centred of the two on randomness as well. Donnelly's own area-and-perimeter form is stable on the same control here — 1.00 ± 0.07 at $n = 50$ tightening to 1.00 ± 0.01 at $n = 2000$ — but since the reference implementation declines to apply it to a polygon we report the `cdf` correction as the external check. Every corrected series falls, but they do not agree on the magnitude: `cdf` steepens the uncorrected series (57.2% against 45.0%), while Donnelly flattens it to 41.9% (0.341 to 0.198), a smaller fall than the raw index rather than a larger one. The direction is robust to the edge treatment; the magnitude is not.

Table S11. Clark–Evans R on the fixed hull under four edge treatments, locations.

Year	Plain	Strictly inside	Border corrected	500 m guard
2013	0.367	0.367	0.350	0.364
2014	0.316	0.316	0.308	0.315
2015	0.280	0.280	0.266	0.274
2016	0.340	0.340	0.312	0.337
2017	0.258	0.258	0.242	0.249
2018	0.441	0.441	0.306	0.359
2019	0.339	0.305	0.281	0.303
2020	0.306	0.255	0.215	0.252
2021	0.429	0.350	0.274	0.326
2022	0.297	0.243	0.189	0.244
2023	0.217	0.196	0.156	0.202
2024	0.204	0.198	0.175	0.199
2025	0.198	0.192	0.177	0.193
2026	0.202	0.196	0.179	0.197

Table S12. Trend in each edge treatment. The four corrected series exclude zero on Newey–West, OLS and Mann–Kendall.

Estimator	Slope/yr	Newey–West interval	OLS p	M–K p
Headline (no correction)	-0.0107	[-0.0188, -0.0027]	0.037	0.010
Strictly inside the hull	-0.0126	[-0.0178, -0.0074]	0.006	0.002
Ripley border correction	-0.0132	[-0.0160, -0.0103]	0.000	0.000
Fixed 500 m guard area	-0.0117	[-0.0152, -0.0082]	0.001	0.000
spatstat `cdf` correction	-0.0159	[-0.0200, -0.0119]	0.000	—

Table S13. The spatstat `cdf`-corrected Clark–Evans series year by year, against the raw index on the same points and window. This is the only series in the paper computed with an independent implementation.

Year	Locations	Raw R	`cdf` R	Difference
2013	39	0.367	0.384	+0.017
2014	97	0.316	0.332	+0.015
2015	227	0.280	0.258	-0.022
2016	186	0.340	0.305	-0.035
2017	518	0.258	0.230	-0.028
2018	354	0.441	0.316	-0.125
2019	458	0.339	0.259	-0.081
2020	675	0.306	0.166	-0.140
2021	729	0.429	0.244	-0.186
2022	527	0.297	0.177	-0.120
2023	464	0.217	0.151	-0.066
2024	499	0.204	0.166	-0.038
2025	546	0.198	0.159	-0.039
2026	561	0.202	0.164	-0.037

Table S14. The 2026 endpoint. 2026 is a nine-month partial year, so every headline quantity is reported with it and with 2025 as the last year.

Series	Slope to 2026	OLS p	Slope to 2025	OLS p	M–K p
Plain R	-0.0107	0.037	-0.0096	0.095	0.030
Inside-hull R	-0.0126	0.006	-0.0125	0.016	0.004
Border R	-0.0132	0.000	-0.0139	0.000	0.001
Guard-area R	-0.0117	0.001	-0.0117	0.003	0.002

S7 THEME COMPOSITION AS A CONFOUNDER

Vehicle share ranges from 58.5% to 73.0% on identifiers across the record and collapses to 16.8% in 2018 on the same basis, so the pooled clustering trend could be a composition effect. Table S15 fits the trend separately within each theme, and the result is that no single theme drives it: the pooled slope is -0.0107 per year, the vehicle theme alone falls at -0.0122 (OLS $p = 0.31$, Mann–Kendall $p = 0.014$), and the themes that grow are small. The composition therefore contributes to the magnitude but does not create the direction.

The last column is what a within-theme null is worth, and it is less than it appears. Each theme is fitted on its own, so each has its own year count and scatter; the minimum detectable effect at 80 % power is larger than the pooled slope for *all seven* themes: the smallest is Air Quality at 0.023 per year and the largest is Sensor Metrics at 0.182, against a pooled slope of 0.011. No theme's own series can resolve an effect of the pooled size, so none of the seven within-theme nulls is evidence of uniformity. The vehicle theme is the instructive case: at 0.035 per year its threshold is three times the pooled slope of 0.011, so its non-significant result is an absence of evidence rather than evidence of absence. This does not weaken the pooled finding, which is carried by the whole record, but it does mean the theme panel cannot support the stronger claim that no subset shows the trend.

Table S15. Clustering trend within each theme, on deduplicated locations. The pooled series is -0.0107 per year for comparison. Minimum detectable effect is at 80 % power on that theme's own year count.

Theme	Slope/yr	OLS p	M–K p	Years	MDE_{80}
People	-0.0251	0.097	0.381	10	0.0427
Environment	-0.0239	0.097	0.083	5	0.0415
Water Level	-0.0203	0.064	0.212	11	0.0301
Vehicles	-0.0122	0.307	0.014	14	0.0349
Air Quality	+0.0110	0.162	0.252	13	0.0226
Sensor Metrics	+0.0216	0.655	0.483	5	0.1818
Weather	+0.0569	0.000	0.000	13	0.0281

S8 GAP-FILLED SENSITIVITY FOR THE MULTISCALE FUNCTION AND THE INFILL TEST

The gap-filled roster is the sensitivity for the published record, since a silent instrument is indistinguishable from a decommissioned one. The main text reports it for R and the concentration index; Table S16 extends it to the multiscale function and the infill test. The multiscale decline is present on the gap-filled roster as well, and the infill count is unchanged at 12 of 13 year pairs above the envelope. The separate exclusion sweep, which drops the discontinuity years rather than filling their gaps, is a different check and does change the count; it is reported in Section S12.

Table S16. The multiscale function at 1 km, published roster against gap-filled roster, with the location count each uses.

Year	n published	n gap-filled	$L(1\,\mathrm{km})$ published	gap-filled
2013	39	39	13640	13640
2014	97	97	7673	7673
2015	227	236	5251	5475
2016	186	281	8991	6573
2017	518	542	6070	6517
2018	354	591	8853	6436
2019	458	533	5716	6647
2020	675	688	7062	6979
2021	729	763	5658	6044
2022	527	560	7445	7657
2023	464	523	5111	5688
2024	499	529	5285	5192
2025	546	560	3962	4047
2026	561	561	4028	4028
Trend on year, 1 km			**-426.4**	**-370.8**
metres per year, published against gap-filled roster				

S9 INFILL DIAGNOSTICS

The homogeneous reference regions are reported here as an alternative to the random-labelling test. All three depend on a homogeneous reference region and so repeat the confound they were meant to remove; they are recorded here for completeness and are not used as evidence.

Table S17. Distance ratios against three homogeneous reference regions, every year pair. A ratio below one reads as infill and above one as expansion; the three do not agree.

Year	New locations	Ratio to box	Ratio to core	Ratio to kernel density
2014	69	0.290	0.571	2.044
2015	146	0.407	1.046	11.013
2016	89	0.091	0.187	2.637
2017	408	0.207	0.526	8.412
2018	156	0.052	0.287	11.557
2019	312	0.164	0.514	12.723
2020	290	0.045	0.234	13.953
2021	140	0.091	0.970	30.702
2022	88	0.035	0.140	5.776
2023	101	0.125	0.277	25.830
2024	125	0.101	0.206	23.136
2025	112	0.170	0.360	39.194
2026	36	0.086	0.185	19.464

Table S18. The random-labelling infill test, all thirteen year pairs.

Year	New locations	Observed median (m)	Permutation envelope (m)	Position
2014	69	1241	[452, 1186]	above
2015	146	1705	[350, 833]	above
2016	89	232	[127, 300]	inside
2017	408	736	[259, 461]	above
2018	156	316	[54, 118]	above
2019	312	646	[106, 235]	above
2020	290	267	[85, 170]	above
2021	140	703	[44, 110]	above
2022	88	140	[30, 85]	above
2023	101	354	[21, 48]	above
2024	125	272	[21, 50]	above
2025	112	402	[20, 48]	above
2026	36	217	[15, 55]	above

S10 THE SNAPSHOT AUDIT IN FULL

The audited snapshot was prepared by the present authors for a separate benchmarking study. Its coverage of the whole census is 64.9%, but that figure mixes two different populations, because the snapshot is a catalogue drawn in 2026 and the census spans every year from 2013. Of the identifiers the two hold in common, only 440 were still reporting in 2026: the rest are retired instruments that the snapshot nevertheless contains, so it is not a current-instrument list. The comparison that isolates the omission is the like-for-like one, against the identifiers active in the snapshot's own year, which is 62.3%.

Table S19. The snapshot against the census, unconditioned and conditioned on the snapshot's own year.

Quantity	Value
Bundle created	2026-09-20
Readings window	2026-05-11 to 2026-05-18
Snapshot identifiers	1411
Identifiers with any reading	24
Census identifiers	1899
Matched	1232
Snapshot-only	179
Census-only	667
Coverage of the census	64.9%
Census identifiers active in 2026	706
of those in the snapshot	440
omitted	266
Coverage conditioned on 2026	62.3%

Table S20. Vehicle contrast, conditioned on identifiers active in the snapshot's own year.

	Vehicles	Non-vehicles	Vehicle share
Omitted (2026-conditioned)	194	72	72.9%
Matched (2026-conditioned)	226	214	51.4%
Odds ratio 2.55; Fisher exact $p = 1.64e\text{-}08$			

S11 THE CLUSTERING CONTROL AS A DRAW, NOT A VALUE

A single complete-spatial-randomness realisation is not a null value: the control is itself a draw, so it carries sampling scatter, and any residual left after averaging is the bias of the estimator rather than noise. Table S21 gives the control at three sample sizes over 199 realisations, and Table S22 separates the two components at each size. The distinction matters because a control that is averaged is often described as sitting on zero when it in fact sits a few metres off it.

Table S21. Border-corrected multiscale L under complete spatial randomness, 199 realisations at each sample size. A negative mean is the estimator's bias, quantified in Table S22.

	$N = 39$		$N = 561$	
Scale	Mean	SD	Mean	SD
100	-86.2	136.5	-20.7	62.6
250	-208.8	233.5	-0.6	46.6
500	-340.3	444.8	0.2	46.0
750	-375.9	638.5	-0.1	48.3
1000	-358.3	768.0	-3.5	50.9
1500	-231.5	866.0	-1.9	45.5
2000	-237.8	872.1	-1.7	50.0

Table S22. The two components of the control at each sample size, pooled over scales: the residual mean absolute bias of the estimator, and the scatter of a single realisation.

	$N = 39$	$N = 200$	$N = 561$
Mean |bias| (m)	262.7	33.3	4.1
Scatter, single realisation (m)	565.6	139.9	50.0

Table S23. Residual bias of the control as a fraction of the observed signal, by scale.

Scale (m)	Bias as share of observed L (%)
100	5.9
250	11.5
500	13.5
750	11.7
1000	9.0
1500	4.2
2000	3.4

Table S24. Leave-one-out intensity: each location's nearest-neighbour distance divided by the distance the leave-one-out kernel density predicts for it ($h = 750$ m, all other points).

Year	N	Median ratio	Share below 1
2013	39	1.27	38.5
2020	675	0.25	76.3
2026	561	0.07	85.4

S12 THE SNAPSHOT'S DUPLICATE REGISTRATIONS AND TWO EXCLUSION SWEEPS

The snapshot holds 179 identifiers absent from the census. Matched by position rather than by identifier, most of them are re-registrations of sites the census already holds, so the raw set difference overstates the disagreement between the two artefacts. Table S25 gives the resolution.

Table S25. The snapshot-only identifiers resolved to the nearest location in the census.

Distance to nearest census location	Identifiers	Share of 179
Within 1 m	41	22.9
Within 5 m	83	46.4
Within 10 m	114	63.7
Within 25 m	141	78.8
Within 50 m	152	84.9
Beyond 50 m	27	15.1

Two further exclusions test whether the headline multiscale and infill results depend on the years with a known publishing discontinuity. Table S26 refits the multiscale trend with the 2017–2018 and 2026 years removed, and reports the infill count on the same subset.

Table S26. Multiscale trend with and without the discontinuity years, on deduplicated locations.

Series	All years	Excluding 2017–2018, 2026
$L(500\,\text{m})$	-313.1	-312.2
$L(1000\,\text{m})$	-426.4	-439.9

Table S27. Infill test: observed medians above the permutation envelope.

	Year pairs above envelope	Year pairs tested
All years	12	13
Excluding 2017–2018, 2026	9	10

The minimum detectable effect for the rarefied concentration slope is 0.0046 per year at 80 % power, against an uncorrected rise of 0.0128 per year.

S13 WHAT THE SNAPSHOT ACTUALLY CONTAINS

The provenance audit counts identifiers, but the two subsets that matter to a user of the snapshot are the ones that carry readings and the ones the census lacks. Neither is a random sample of the catalogue. The snapshot stores no theme field, so `broker_name` is used as the publisher-side proxy: it names the feed an instrument is ingested from, which is what the archive's own theme directories are built from.

Table S28. The identifiers carrying readings, by ingestion feed. The catalogue spans 27 feeds.

Feed (broker)	Identifiers	Share of the 24 with readings
aq_mesh_api	16	66.7
LORA	8	33.3

Table S29. Feeds contributing identifiers the census lacks, against the same feeds' share of the matched set. Shares within each column are rounded to one decimal and need not sum to 100.

	Snapshot-only (179)		Matched (1232)	
Feed (broker)	*n*	%	*n*	%
People Counting API	78	43.6	407	33.0
Internal Building Sensors	47	26.3	3	0.2
Building External Sensor	26	14.5	39	3.2
Wunderground API	9	5.0	51	4.1
Newcastle Council RSS feed	7	3.9	88	7.1
Car Park	3	1.7	24	1.9
Emote Air Quality Sensor	3	1.7	134	10.9
NE Travel Data API	3	1.7	310	25.2

Two things follow. The reading-bearing subset is thematically narrow: it is dominated by air quality and the LoRa network, and its readings are the air-quality and weather variables rather than the vehicles, people counters or travel data that make up most of the catalogue, so an execution target built on it exercises one corner of the platform. And the snapshot-only subset is concentrated in building instrumentation, which is the opposite direction from the omission contrast in Section S10: there the snapshot is short of vehicle counters, here it is long on building sensors. Both are descriptions of a frozen artefact, not findings about the observatory, and neither can be tested against a base rate outside this platform.

An implementation note, because it is a silent failure mode and not a diagnostic message. The readings table joins to the sensor table on the sensor *name*, not on the numeric `raw_id`, while the census is keyed on `raw_id`. Joining the readings table on `raw_id` yields an empty intersection (0 identifiers) and makes it look as though none of the reading-bearing identifiers are in the census; the correct join gives 24, of which 24 are present.

S14 CORE-DEFINITION SENSITIVITY

The core panel of the frame table uses a symmetric 5–95% rectangle—percentile trims on $|x|$ and $|y|$ about the projection origin, not an interquartile box—which is an arbitrary support. Table S30 sweeps it over five trims and adds four non-rectangular cores, so the check is not confined to one family of supports. Every row declines and every row is significant, so the core panel reports the network rather than the percentile.

Table S30. Clark–Evans R trend on nine core definitions, deduplicated locations, each with its own support as the density denominator.

Core	R_{2013}	R_{2026}	Slope/yr	OLS p
Trim 1st–99th pct	0.785	0.301	-0.0298	< 0.001
Trim 2.5th–97.5th pct	0.917	0.261	-0.0427	< 0.001
Trim 5th–95th pct (headline)	0.812	0.251	-0.0368	< 0.001
Trim 10th–90th pct	0.661	0.250	-0.0342	< 0.001
Trim 25th–75th pct	0.807	0.283	-0.0383	< 0.001
Convex hull, central 80%	1.056	0.275	-0.0527	< 0.001
Fixed 2 km disc	0.472	0.419	-0.0219	0.002
Fixed 3 km disc	0.314	0.332	-0.0185	0.007
Fixed 5 km disc	0.592	0.248	-0.0278	< 0.001

S15 GRID-ORIGIN SENSITIVITY FOR THE CONCENTRATION INDEX

The concentration index is computed on a grid fixed in geographic space, so its origin is a choice. Table S15 shifts it over a 5×5 lattice of sub-cell positions at the headline resolution and at a finer one. Every origin shows a rise, so the sign of the uncorrected trend is not an artefact of where the grid starts; the magnitude varies by about 40% at the coarser resolution and by about 90% at the finer one, which is why the text quotes the index only with its partition named.

The same script re-checks Equation (1) of the main text against the textbook form of the Gini coefficient over 2000 random count vectors. The maximum absolute difference is 4.4×10^{-16}, so the printed formula and the code agree to machine precision.

Table S31. Concentration-index trend over 25 grid origins, at two resolutions.

Cells per side	Slope range	Mean	SD	All origins rise
20	0.0120–0.0167	0.0141	0.0016	yes
40	0.0072–0.0138	0.0103	0.0017	yes

S16 INDEPENDENT CROSS-CHECK AGAINST SPATSTAT

The multiscale curve is computed with our own code. Table S32 compares it with spatstat 3.8.3 under R 4.6.1, on the same points with the same polygonal window and spatstat's own border correction. The two agree to a median of 0.18 per cent, with the largest disagreement (6.09 per cent) at 2 km, where the eroded window retains the fewest pairs.

One convention difference is worth recording so that a reader comparing the two implementations does not mistake it for a disagreement: spatstat's `Lest` column named `border` holds $\sqrt{K/\pi}$, whereas this paper's L is $\sqrt{K/\pi} - r$. The two therefore differ by exactly r by construction, and the comparison here adds r back so that like is compared with like.

Table S32. Manuscript estimator against spatstat `Lest` (border correction), deduplicated locations, by scale. Absolute relative difference over the fourteen years.

Scale (m)	Median diff (%)	Max diff (%)
100	0.194	0.538
250	0.125	1.375
500	0.108	0.293
750	0.002	1.202
1000	0.228	1.231
1500	0.247	1.021
2000	1.414	6.088

S17 THINNING INVARIANCE: DO THE TWO INDICES MEASURE THE SAME THING?

K and L are invariant under independent random thinning of a fixed pattern; the Clark–Evans index is not written that way. Table S33 gives each year's observed R against the R of a size-matched random draw from the pooled set of 1567 distinct locations, which is the value each year's *size alone* predicts for this network's structure. A negative excess means the year is *more* clustered than its size predicts, because R falls as clustering rises.

The 2013 network is indistinguishable from its size-matched draw; the 2026 network is more clustered than its own size predicts by 7.0 standard deviations. On a fixed clustered pattern that is never moved, R itself changes by only 11 per cent across a sixteen-fold change in n, so the mechanism in which a falling R is mechanical is regime-dependent rather than automatic, and it has to be settled on the data.

The two references above disagree, and the disagreement is the subject of Section S18: thinning the 2026 network to $n = 39$ takes R from 0.202 to 0.387, so size alone reproduces the whole decline, whereas the pooled reference is nearly flat in n (0.449 to 0.436). The near-duplicate test shows that the decline is substantially a co-located-registration effect, which thinning and pooling absorb but the published exact-coordinate deduplication does not.

Table S33. Observed R against the 2026 network thinned to each year's size. The $n = 561$ row is zero by construction.

Year	n	R observed	R from thinned 2026	Excess
2013	39	0.367	0.387	-0.019
2014	97	0.316	0.384	-0.068
2015	227	0.280	0.350	-0.070
2016	186	0.340	0.357	-0.017
2017	518	0.258	0.222	+0.037
2018	354	0.441	0.294	+0.148
2019	458	0.339	0.250	+0.090
2020	675	0.306	0.202	+0.104
2021	729	0.429	0.202	+0.228
2022	527	0.297	0.218	+0.079
2023	464	0.217	0.248	-0.031
2024	499	0.204	0.231	-0.027
2025	546	0.198	0.209	-0.011
2026	561	0.202	0.202	+0.000

Table S34. Observed R against a size-matched random draw from the pooled structure, 400 draws per year. Negative z means more clustered than the reference.

Year	n	R observed	R from the pool	SD	z
2013	39	0.367	0.449	0.104	-0.78
2014	97	0.316	0.454	0.081	-1.70
2015	227	0.280	0.458	0.060	-2.99
2016	186	0.340	0.462	0.063	-1.93
2017	518	0.258	0.439	0.038	-4.77
2018	354	0.441	0.448	0.048	-0.15
2019	458	0.339	0.443	0.041	-2.53
2020	675	0.306	0.428	0.032	-3.86
2021	729	0.429	0.425	0.029	+0.16
2022	527	0.297	0.440	0.038	-3.79
2023	464	0.217	0.441	0.039	-5.67
2024	499	0.204	0.439	0.037	-6.36
2025	546	0.198	0.439	0.036	-6.67
2026	561	0.202	0.436	0.033	-7.00

S18 NEAR-DUPLICATE REGISTRATIONS: THE HEADLINE DECLINE IS TOLERANCE-DEPENDENT

The exact-coordinate deduplication used throughout the paper collapses only identifiers at the *same* coordinate. Two registrations a few metres apart survive it as two "locations". Table S36 reclusters every year's identifiers by single-linkage at radius r, keeping one identifier per cluster, and recomputes the whole fixed-hull series. The exact-coordinate row reproduces the published series, which is the control on the method.

The decline is not robust to the tolerance. At 25 m the fall is 4.3% and the fitted slope is $+0.0014$ per year ($p = 0.74$), against 45.3% and $p = 0.035$ on the published deduplication. Table S35 sweeps the radius from 1 m to 50 m and shows the fall to be monotone with no plateau, reversing sign by 50 m; the radius is therefore a stated choice and not a threshold the data identify. The merged identifiers are concentrated in the recent years: the 25 m merge removes 306 of 2026's 706 identifiers, 145 of which the exact-coordinate deduplication had already removed, against 2 of 2013's 41. They are co-located instruments at one installation rather than dispersed events, so at large n they depress the nearest-neighbour mean, and thinning or pooling breaks them up. That is the reconciliation of the two size-matched references of Section S17, which disagree because the pool has already absorbed the pairs and the thinned reference has broken them.

Table S35. The merge tolerance swept from 1 m to 50 m, single-linkage with one identifier kept per cluster. The fall has no plateau, and it reverses sign by 50 m.

Merge radius	2013 R	2026 R	Fall	OLS p
1 m	0.367	0.202	44.9%	0.037
2 m	0.367	0.203	44.7%	0.038
5 m	0.367	0.211	42.5%	0.053
10 m	0.367	0.242	34.1%	0.120
15 m	0.367	0.290	21.1%	0.359
25 m	0.367	0.351	4.3%	0.742
50 m	0.367	0.392	-6.8%	0.320

The same result is reported in the main text's tolerance figure, which also gives the nearest-neighbour distance distribution. That distribution is smooth: the largest gap below 200 m is 35 m in 2014 and 6 m to 26 m in every other year, so no trough separates a co-located population from distinct installations and no radius is singled out by the data. The merge is also not specific to single-linkage: complete-linkage at the same radius gives 4.7% ($p = 0.77$) and a centroid representative gives 4.3%, while the largest cluster single-linkage forms at 25 m has a diameter of 74 m against a median of 12 m, so chaining is bounded. Snapping to a 25 m grid instead gives 31.1% ($p = 0.15$), which shows the result is not entirely independent of the merge rule.

Table S36. The fixed-hull location series under tolerance deduplication. Clusters are formed by single-linkage at radius r and one identifier is kept per cluster. The exact-coordinate row is the published series.

Deduplication	2013 R	2026 R	Fall	Slope/yr	OLS p	2026 pooled z
Exact coordinate (published)	0.367	0.201	45.3%	-0.0108	0.035	-6.6
Within 5 m	0.367	0.211	42.5%	-0.0101	0.053	-6.8
Within 10 m	0.367	0.242	34.1%	-0.0076	0.120	-6.0
Within 25 m	0.367	0.351	4.3%	+0.0014	0.742	-3.6

Table S37. Identifiers merged by tolerance deduplication, by year. The tight clusters accumulate over the record.

Year	Identifiers	Sites at 5 m	Sites at 10 m	Sites at 25 m	Removed at 25 m
2013	41	39	39	39	2
2014	97	95	93	90	7
2015	228	222	215	193	35
2016	199	180	178	171	28
2017	605	504	478	417	188
2018	367	340	328	310	57
2019	474	444	425	360	114
2020	749	652	622	532	217
2021	800	699	672	575	225
2022	566	499	465	379	187
2023	466	442	409	329	137
2024	549	482	435	335	214
2025	707	530	494	388	319
2026	706	543	505	400	306

Table S38. The pooled size-matched reference recomputed on the 25 m deduplication, so Table S34 can be re-read under the wider tolerance. The separation shrinks but does not vanish.

Year	n	R observed	R from pool	SD of draws	z
2013	39	0.367	0.498	0.111	-1.18
2014	90	0.331	0.510	0.083	-2.16
2015	193	0.348	0.511	0.059	-2.75
2016	171	0.357	0.512	0.064	-2.40
2017	417	0.345	0.507	0.042	-3.85
2018	310	0.485	0.510	0.047	-0.51
2019	360	0.452	0.504	0.045	-1.16
2020	532	0.381	0.500	0.034	-3.51
2021	575	0.523	0.499	0.032	+0.74
2022	379	0.434	0.510	0.045	-1.70
2023	329	0.361	0.508	0.047	-3.15
2024	335	0.363	0.504	0.048	-2.96
2025	388	0.348	0.510	0.044	-3.66
2026	400	0.351	0.505	0.042	-3.64

S19 MULTIPLICITY, THE RANDOM SEED AND THE CODE

The paper reports about 182 fitted comparisons when the level series are counted by frame, unit and year, the concentration index by resolution and year, the rarefaction anchors, the interval procedures and the bandwidth sweep. They are not corrected as one family, because many are descriptive readings of one series under different frames rather than independent hypotheses, but the size of the search is stated.

The one family that *is* corrected as a family is the per-year clustering test, because those tests are the same question asked repeatedly. Table S39 gives each year pair its raw one-sided Monte Carlo p-value and the Benjamini–Hochberg adjusted value across the thirteen pairs. The correction is applied to thirteen tests rather than fourteen years, because each pair is compared only against the pooled positions of that pair; it is approximate, because successive pairs share a year. 10 of the 13 year pairs survive at a 5 % false-discovery rate: 2014, 2015, 2016, 2017, 2019, 2022, 2023, 2024, 2025, 2026. The three that do not — 2018, 2020, 2021 — are the same years whose observed distance falls inside the pooled-subset interval at the single-test level, so the correction removes exactly the years the uncorrected test could not separate.

Table S39. Per-year clustering test against the pooled-subset null, with the Benjamini–Hochberg step-up. The p-value is one-sided: the share of pooled random subsets with a median new-to-new distance no greater than the observed one.

Year	n new	Obs. (m)	Pooled 95% (m)	p	p_{BH}
2014	69	255	[272, 816]	0.012	0.017
2015	146	48	[190, 472]	0.000	0.000
2016	89	210	[299, 796]	0.000	0.000
2017	408	39	[82, 140]	0.000	0.000
2018	156	327	[230, 658]	0.188	0.223
2019	312	26	[122, 287]	0.000	0.000
2020	290	227	[165, 307]	0.521	0.564
2021	140	509	[304, 653]	0.679	0.679
2022	88	15	[374, 931]	0.000	0.000
2023	101	17	[278, 770]	0.000	0.000
2024	125	20	[303, 696]	0.000	0.000
2025	112	60	[411, 775]	0.000	0.000
2026	36	722	[760, 1897]	0.020	0.026

All draws use one fixed seed (`20261003`). The seed enters the rarefied series through the subsampling draws, and Table S40 bounds its effect.

Table S40. Random-seed sensitivity of the rarefied clustering slope on the fixed hull.

Seed	Slope per year	Interval contains zero
20261003	+0.0003	yes
20261004	+0.0003	yes
20261005	+0.0005	yes

Table S41. Analysis scripts and the files they produce.

Script	Produces
`build_archive_index.py`	`uo_archive_index.json`, the per-year census
`analysis_common.py`	core analyses, reference frames, uncertainty
`analysis_multiscale.py`	multiscale estimator and its controls
`analysis_trendtests.py`	correctly sized trend tests, infill count
`analysis_inhomogeneous.py`	multiscale vs the inhomogeneous null
`analysis_snapshot.py`	the snapshot audit, conditioned
`analysis_edge.py`	edge correction, endpoint, themes, gap-filled
`analysis_envelope.py`	CSR envelope, leave-one-out intensity
`analysis_exclusions.py`	Gini MDE, exclusion sweeps
`analysis_snapdupes.py`	snapshot-only identifiers, by location
`analysis_feeds.py`	snapshot composition by ingestion feed
`analysis_coredefs.py`	core-definition (MAUP) sweep
`analysis_gridorigin.py`	grid-origin sweep, equation check
`analysis_donnelly.py`	Donnelly edge correction
`analysis_thinning.py`	thinning invariance, size-matched test
`analysis_neardup.py`	tolerance deduplication, the near-duplicate test
`crosscheck_spatstat.R`	independent comparison against spatstat
`crosscheck_spatstat.py`	compares the two estimators
`crosscheck_donnelly.R`	Clark–Evans against spatstat's cdf correction
`make_study_area_figure.py`	Figure 1, the study area
`make_supplement.py`	this document
`verify_ripley_sign.py`	known-answer control for the multiscale estimator

REFERENCES

REFERENCES

Baddeley, A. and Turner, R. (2005). spatstat: An R package for analyzing spatial point patterns. *Journal of Statistical Software*, 12(6).

Donnelly, K. P. (1978). Simulations to determine the variance and edge effect of total nearest-neighbour distance. In Hodder, I., editor, *Simulation Studies in Archaeology*, pages 91–95. Cambridge University Press, Cambridge.

Ripley, B. D. (1977). Modelling spatial patterns. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 39(2):172–192.

Generative AI Use Disclosure Statement

Tool and version. GitHub Copilot 0.69.0, used to support LaTeX preparation and language editing of this supplement. It did not generate or alter any tabulated value; every table is produced by the analysis scripts in the deposit. **Originality and accuracy.** The authors confirm the originality and accuracy of the content. **Terms of use.** The authors have checked the tool's terms of use and confirm its use is suitable for publication. **Responsibility.** The authors take full responsibility for the integrity of the whole content, including the accuracy of all references.